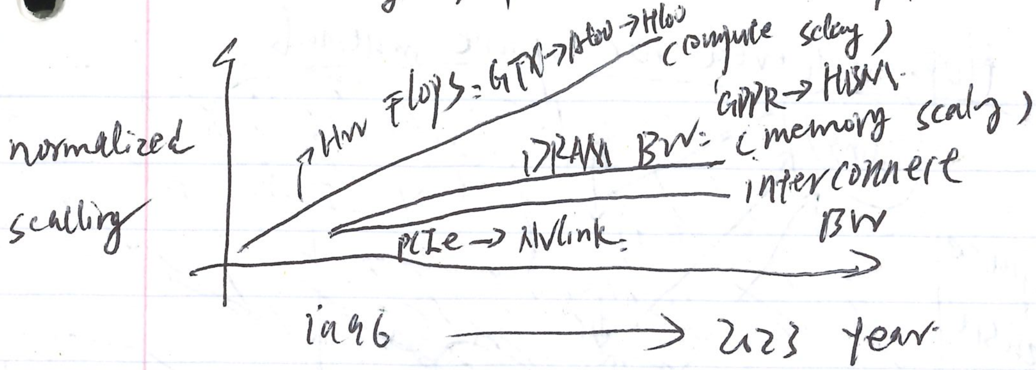
Compute scaling is faster than memory scaling

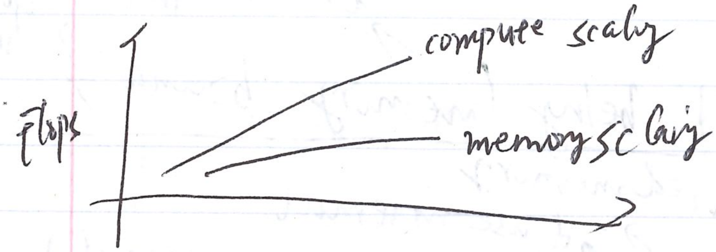
scaling of peak hardware flops, and memory/inter connect Bandwidth.



(very nice chart)

PCIe → NvLink: ~~GPU connect~~ connect GPU to host/server.

NvSwitch: connect GPU to GPU



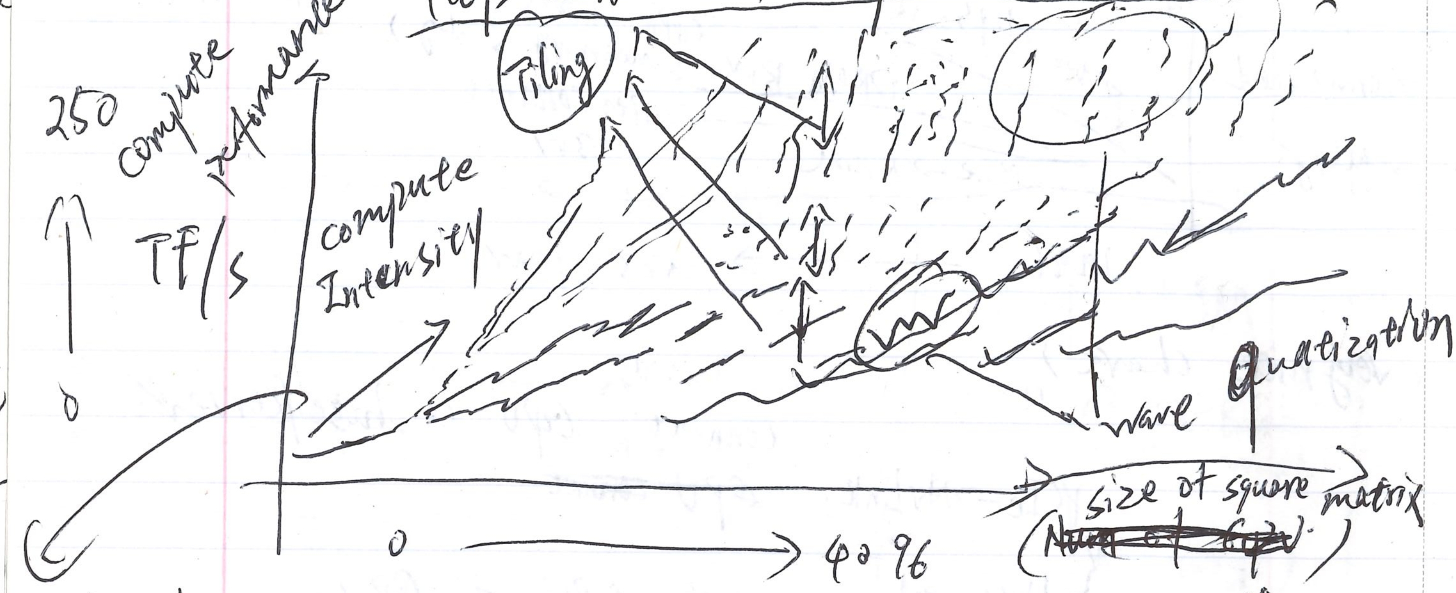
II) GPU summary:

GPUs are massively parallel — same instructions applied across many workers

compute (and esp matmuls) have scaled faster than memory.

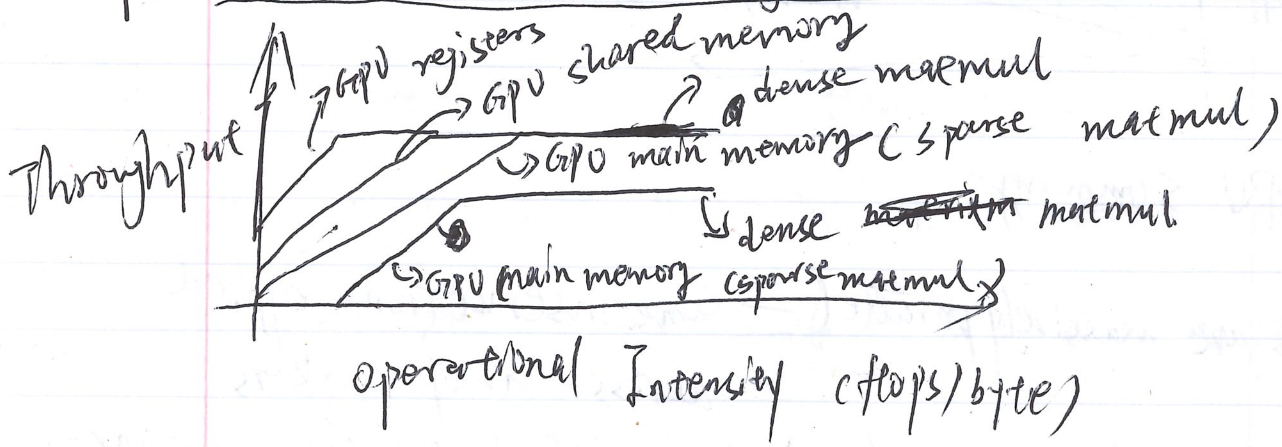
we have to respect the memory hiercely to make things so fast part #2: make ML workloads fast on GPU

Flops achieved for square matmuls



each point: results of matrix multiplication on GPU

Key: how do we avoid being (memory bound)? → throughput doesn't increase when operation increases



Solutions to make workloads run fast on GPU

#1: Control Divergence (not a memory issue)

GPU operates in a SIMT (one instruction for multiple threads) model.

↳ every thread in a warp is executing the same instruction